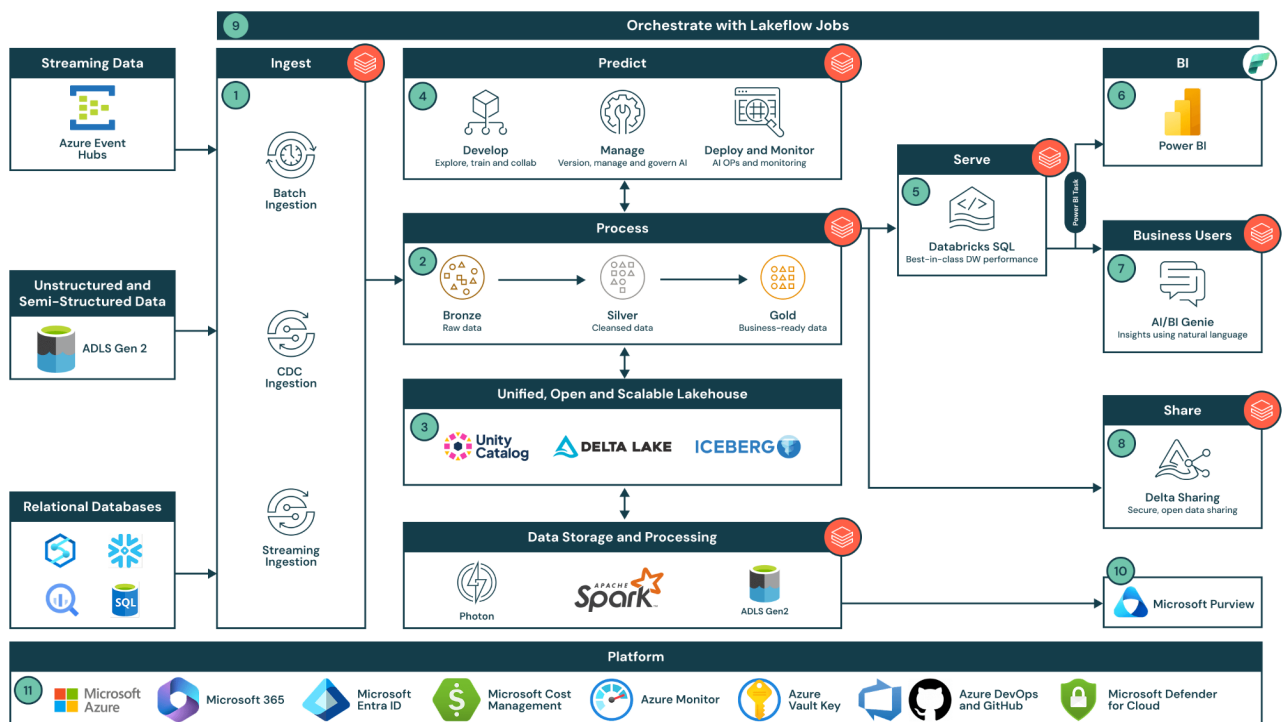
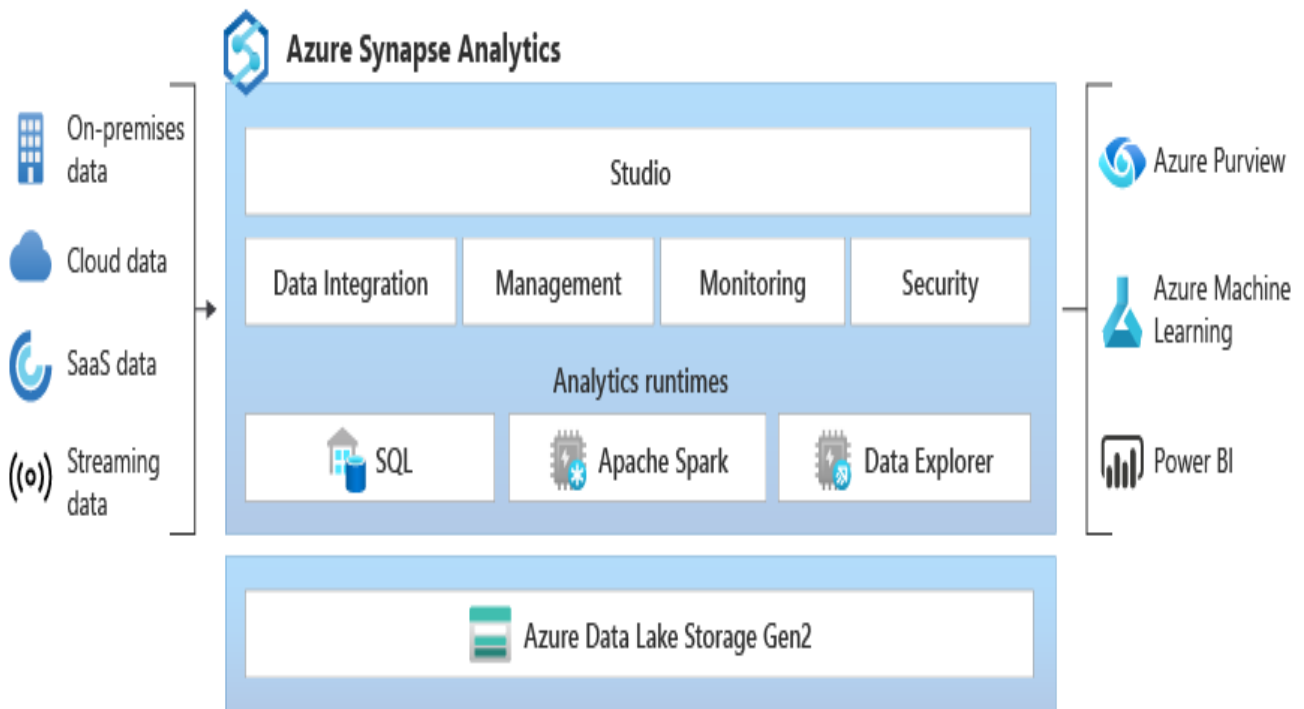


Synapse and Databricks Architecture Differences



Azure Synapse vs Azure Databricks

Azure Synapse is an all-in-one analytics platform focused on SQL-based data warehousing and analytics.

Azure Databricks is a Lakehouse platform focused on Apache Spark, Delta Lake, Data Engineering, AI, and Machine Learning.

Architecture Philosophy

Azure Synapse: Integrated Analytics Platform, SQL-first, combines SQL Pools, Spark Pools, Pipelines, Monitoring and Security in one workspace.

Azure Databricks: Spark-first Lakehouse platform built around Delta Lake, Photon, Unity Catalog, Lakeflow Jobs and Databricks SQL.

Key Differences

- Platform Type: Synapse = Analytics Platform | Databricks = Lakehouse Platform
- Architecture: SQL-first | Spark-first
- Purpose: Data Warehouse & BI | Data Engineering & AI
- Primary Engine: SQL | Apache Spark
- Best For: Reporting | Big Data Processing

1. Purpose

Synapse: Enterprise Data Warehouse, BI, SQL Analytics, Reporting, ETL. Users: SQL Developers, Analysts.

Databricks: Data Engineering, Big Data, ML, AI, Streaming. Users: Data Engineers, Data Scientists.

2. Storage

Synapse: ADLS Gen2 + Dedicated SQL Pool. Data is often copied into the warehouse.

Databricks: ADLS Gen2 + Delta Lake. Data is stored as Delta Tables using Bronze → Silver → Gold.

3. Compute Engines

Synapse: Dedicated SQL Pool (MPP), Serverless SQL Pool (pay per TB scanned), Apache Spark Pool.

Databricks: Apache Spark + Photon Engine for faster Spark and SQL execution.

4. Data Processing

Synapse: SQL, Spark, Data Flows (SQL-centric).

Databricks: Spark SQL, DataFrames, Delta Lake, Streaming, AI (Spark-centric).

5. ETL & Orchestration

Synapse: Synapse Pipelines (similar to Azure Data Factory).

Databricks: Lakeflow Jobs with notebook-based workflows.

6. Data Warehouse

Synapse: Traditional Enterprise Data Warehouse with Star/Snowflake schemas.

Databricks: Databricks SQL querying Delta Tables (no traditional warehouse).

7. Data Lake Architecture

Synapse: Landing → Raw → Curated → Warehouse.

Databricks: Bronze → Silver → Gold (Medallion Architecture).

8. Metadata & Governance

Synapse: Hive Metastore, Azure Purview.

Databricks: Unity Catalog for metadata, governance, lineage and access control.

9. Security

Synapse: Azure AD, RBAC, SQL Permissions, Managed Identity.

Databricks: Unity Catalog, Azure AD, Row-level security, Column masking.

10. Machine Learning

Synapse: Azure Machine Learning integration.

Databricks: MLflow, AutoML, Feature Store, Model Registry, Mosaic AI.

11. Streaming

Synapse: Basic streaming through Spark.

Databricks: Structured Streaming, Auto Loader, Kafka, Event Hubs.

12. Performance

Synapse: Excellent SQL performance.

Databricks: Excellent Spark performance with Photon optimizations.

13. Cost

Synapse: Separate charges for SQL Pools, Spark Pools and Pipelines.

Databricks: Compute (DBUs) + Storage with autoscaling and auto-termination.

14. SQL Experience

Synapse: Best for T-SQL developers.

Databricks: Databricks SQL optimized for Delta Lake.

15. AI & Modern Data Platform

Synapse: Azure ML integration.

Databricks: AI Agents, Vector Search, MLflow, Mosaic AI, LLM support, Model Serving.

When to Use Azure Synapse

- Traditional Enterprise Data Warehouse
- Heavy SQL workloads
- Existing SQL Server ecosystem
- Power BI reporting
- Star Schema implementations

When to Use Azure Databricks

- Large-scale ETL
- Lakehouse architecture
- Streaming pipelines
- Machine Learning & AI

- Bronze–Silver–Gold architecture

Real-World Use Cases

- Enterprise Data Warehouse → Synapse
- SQL Reporting → Synapse
- Power BI Dashboards → Synapse
- SQL Server Migration → Synapse
- Large-scale ETL → Databricks
- Bronze-Silver-Gold Pipelines → Databricks
- Streaming → Databricks
- Data Engineering → Databricks
- Machine Learning → Databricks
- AI & LLM Applications → Databricks